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Exact theory for superconductivity in a doped Mott insulator

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SUPPLEMENTARY MATERIALS

Pair binding instability with zero temperature Hatsugai-Kohmoto Gibbs state

Given the Gibbs state $\rho = e^{-\beta H_{\text{HK}}}/Z = \sum_n e^{-\beta E_n} |n\rangle\langle n|$, we fix the purification on the doubled Hilbert space with $\mathcal{H}_B \simeq \mathcal{H}_A$

$$|\beta\rangle = \sum_n e^{-\beta E_n/2} |n\rangle \otimes |n\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B \quad (35)$$

satisfying $\text{tr}_B |\beta\rangle\langle\beta| = \rho$, and restrict

$$A^\dagger = \sum_{k \notin \Omega_2} \alpha_k b_k^\dagger \quad (36)$$

to the singly-occupied and unoccupied regions Ω_1 and Ω_0 of the Brillouin zone. Now $|\psi\rangle = A^\dagger \otimes \mathbb{1} |\infty\rangle$ has overlap

$$\langle\infty|b_k \otimes \mathbb{1}|\psi\rangle = \text{tr}_{A,B} |\infty\rangle\langle\infty| (b_k A^\dagger \otimes \mathbb{1}) \quad (37)$$

$$= \text{tr}_A \rho b_k A^\dagger = \langle b_k A^\dagger \rangle \quad (38)$$

$$= \begin{cases} 0 & \text{if } k \in \Omega_2, \\ \frac{1}{4}\alpha_k & \text{if } k \in \Omega_1, \\ \alpha_k & \text{if } k \in \Omega_0. \end{cases} \quad (39)$$

With $[b_k, H]$ as before, for each $k \in \Omega_1, \Omega_0$ we have

$$\begin{aligned} & C_k i\hbar \partial_t \alpha_k(t=0) \\ &= \langle\infty| e^{i(H \otimes \mathbb{1} + \mathbb{1} \otimes H')t} (b_k \otimes \mathbb{1}) e^{-i(H \otimes \mathbb{1} + \mathbb{1} \otimes H')t} |\psi\rangle \Big|_{t=0} \end{aligned} \quad (40)$$

$$= \langle\infty| [b_k, H] \otimes \mathbb{1} |\psi\rangle \quad (41)$$

$$\begin{aligned} &= (2\xi_k + U \langle n_{k\downarrow} + n_{-k\uparrow} \rangle) C_k \alpha_k \\ &\quad - \frac{g}{L^d} \langle 1 - n_{k\uparrow} - n_{-k\downarrow} \rangle \left[\sum_{q \in \Omega_1} \frac{1}{4} \alpha_q + \sum_{q \in \Omega_0} \alpha_q \right] \end{aligned} \quad (42)$$

where $C_k = 1/4$ if $k \in \Omega_1$ and $C_k = 1$ if $k \in \Omega_0$. Then taking $\alpha_k(t) = e^{-iEt/\hbar} \alpha_k(0)$ recovers the same consistency equation

$$1 = -\frac{g}{L^d} \sum_{k \in \Omega_1, \Omega_0} \frac{\langle 1 - n_{k\uparrow} - n_{-k\downarrow} \rangle}{E - 2\xi_k - U \langle n_{k\downarrow} + n_{-k\uparrow} \rangle} \quad (43)$$

$$= -g \int_{\mu}^{W/2} d\epsilon \frac{\rho(\epsilon)}{E - 2\epsilon + 2\mu} \quad (44)$$

since the numerator vanishes in the singly-occupied region Ω_1 .

Dyson equation for pair susceptibility

In order to relate the pair susceptibility

$$\chi(i\nu_n) \equiv \frac{1}{L^d} \int_0^\beta d\tau e^{i\nu_n \tau} \langle T \Delta(\tau) \Delta^\dagger \rangle_g \quad (45)$$

to the bare pair susceptibility, $\chi_0(i\nu_n)$ at $g = 0$, we work in the interaction picture where

$$\langle T \Delta(\tau) \Delta^\dagger \rangle_g = \frac{\langle TS(\beta, 0) \Delta(\tau) \Delta^\dagger \rangle_0}{\langle TS(\beta, 0) \rangle_0} \quad (46)$$

for $\Delta(\tau)$ evolved in the Heisenberg picture (under H) in $\langle \cdots \rangle_g$, and evolved in the interaction picture (under H_{HK}) in $\langle \cdots \rangle_0$. Here

$$S(\beta, 0) = e^{\beta H_{\text{HK}}} e^{-\beta H} = T_{\tau_i} \exp g \int_0^\beta d\tau_1 H_p(\tau_1), \quad (47)$$

for $H_p(\tau_1)$ evolved in the interaction picture. Then the numerator takes the form of a power series in g ,

$$\begin{aligned} & \langle TS(\beta, 0) \Delta(\tau) \Delta^\dagger \rangle_0 \\ &= \sum_{m=0}^{\infty} \frac{g^m}{m!} \langle T \left(\int_0^\beta d\tau_1 H_p(\tau_1) \right)^m \Delta(\tau) \Delta^\dagger \rangle_0 \end{aligned} \quad (48)$$

where each term

$$\frac{1}{L^d} \langle T \left(\int_0^\beta d\tau_1 H_p(\tau_1) \right)^m \Delta(\tau) \Delta^\dagger \rangle_0 = \frac{1}{(L^d)^{m+1}} \int \cdots \int_0^\beta d\tau_1 \cdots d\tau_m \langle T \Delta^\dagger(\tau_1) \Delta(\tau_1) \cdots \Delta^\dagger(\tau_m) \Delta(\tau_m) \Delta(\tau) \Delta^\dagger \rangle_0 \quad (49)$$

factorizes, as χ_0 does in Eq. (20), because each pair annihilator $b_k(\tau)$ in the sum

$$\Delta(\tau) = \sum_k b_k(\tau) = \sum_k e^{-\tau(2\xi_k + U(n_{k\downarrow} + n_{-k\uparrow} - 1))} b_k \quad (50)$$

evolves (in the interaction picture under H_{HK}) as a multiple of an unevolved pair annihilator b_k . The denominator then removes all disconnected factorizations with

either $\langle T \Delta^\dagger(\tau_i) \Delta(\tau_i) \rangle_0$ for any time τ_i or $\langle T \Delta(\tau) \Delta^\dagger \rangle_0$. This leaves only those factorizations with the form of an m -wise convolution, resulting in

$$\chi(i\nu_n) = \sum_{m=1}^{\infty} g^{m-1} (\chi_0(i\nu_n))^m = \frac{\chi_0(i\nu_n)}{1 - g\chi_0(i\nu_n)}. \quad (51)$$

Example of T_c and Δ calculation

To ease calculations, consider ϵ_k such that $\rho(\omega) = \frac{1}{L^d} \sum_k \delta(\omega - \epsilon_k) = \frac{1}{W}$ for $-\frac{W}{2} < \omega < \frac{W}{2}$. We will focus on the half-filled metal i.e. $U < W$ and $\mu = U/2$.

The single-particle density of states (DOS) can be broken into contributions from different regions of momentum space

$$N(\omega) = N_0(\omega) + N_2(\omega) + \frac{1}{2}N_1(\omega) \quad (52)$$

$$N_0(\omega) = \theta(\omega)\rho(\omega + U/2) \quad (53)$$

$$N_2(\omega) = \theta(-\omega)\rho(\omega - U/2) \quad (54)$$

$$N_1(\omega) = \theta(-\omega)\theta(\omega + U)\rho(\omega + U/2) \quad (55)$$

$$+ \theta(\omega)\theta(-\omega + U)\rho(\omega - U/2) \quad (56)$$

The effective DOS for calculating T_c and Δ are

$$N'(\omega) = N_0(\omega) + N_2(\omega) + \frac{1}{4}N_1(\omega) \quad (57)$$

$$N''(\omega) = N_0(\omega) + N_2(\omega) + N_1(\omega) \quad (58)$$

$$(59)$$

a. Superconducting temperature T_c . The susceptibility diverges when

$$\frac{1}{g} = \chi_0(0) = \int d\omega N'(\omega) \frac{\tanh \frac{\beta\omega}{2}}{2\omega}. \quad (60)$$

Set $x = \beta\omega/2$ and integrate by parts

$$\frac{1}{g} = -\frac{1}{2} \int dx \ln x \left[N' \left(\frac{2x}{\beta} \right) \text{sech}^2 x \right. \quad (61)$$

$$\left. + \left(\frac{d}{dx} N' \left(\frac{2x}{\beta} \right) \right) \tanh x \right]. \quad (62)$$

For $T \ll U, W$ this becomes

$$\frac{1}{g} = \frac{1}{W} \ln \frac{\beta(W-U)}{4} + \frac{1/4}{W} \ln \frac{\beta U}{4} \quad (63)$$

$$- N'(0) \left(-\ln \left(\frac{4}{\pi} \right) - \gamma \right) \quad (64)$$

$$\frac{W}{g} = \ln \frac{\beta(W-U)}{4} + \frac{1}{4} \ln \frac{\beta U}{4} - \frac{5}{4} \left(-\ln \left(\frac{4}{\pi} \right) - \gamma \right) \quad (65)$$

where $\gamma \approx 0.577$ is Euler's constant. The solution gives the transition temperature

$$T_c = (W-U)^{4/5} U^{1/5} \frac{e^\gamma}{\pi} e^{-\frac{4}{5} \frac{W}{g}}. \quad (66)$$

b. Superconducting gap Δ . The gap equation is given by

$$1 = \frac{g}{2} \int d\omega \frac{N''(\omega)}{\sqrt{\omega^2 + |\Delta|^2}} \quad (67)$$

$$= \frac{g}{W} \sinh^{-1} \left(\frac{W-U}{2\Delta} \right) + \frac{\alpha g}{W} \sinh^{-1} \left(\frac{U}{2\Delta} \right) \quad (68)$$

For $\Delta \ll U, W$ this becomes

$$\frac{1}{g} = \frac{1}{W} \ln \left(\frac{W-U}{\Delta} \right) + \frac{\alpha}{W} \ln \left(\frac{U}{\Delta} \right) \quad (69)$$

which can be solved to find

$$\Delta = (W-U)^{1/2} U^{1/2} e^{-\frac{W}{2g}} \quad (70)$$

Variational ground state

Consider the variational wave function

$$|\psi\rangle = \prod_{k>0} \left(x_k + y_k b_k^\dagger b_{-k}^\dagger + \frac{z_k}{\sqrt{2}} (b_k^\dagger + b_{-k}^\dagger) \right) |0\rangle. \quad (71)$$

$\langle\psi|\psi\rangle = 1$ is satisfied if $|x_k|^2 + |y_k|^2 + |z_k|^2 = 1$. This generalizes the BCS wavefunction, which corresponds to $x_k = u_k^2$, $y_k = v_k^2$, $z_k = \sqrt{2}u_k v_k$. Furthermore, the state defined by $x_k = 1$ for $k \in \Omega_0$, $z_k = 1$ for $k \in \Omega_1$, and $y_k = 1$ for $k \in \Omega_2$ is a ground state of the HK model. Note that although one signal of pair condensation in the BCS wavefunction is the presence of nonzero $u_k v_k \propto z_k$, this state is not a pair condensate.

In the free fermion case, the ground state in the absence of pairing is the filled Fermi sea, with $u_k = 1$ for $k \in \Omega_0$ and $v_k = 1$ for $k \in \Omega_2$. For a small pairing interaction g , the variational ground state with pairing is very similar but with both u_k and v_k non-zero near the boundary of Ω_0 and Ω_2 , namely the Fermi surface. In the HK model with weak pairing ($g \ll U, W$), we similarly expect that both x_k and z_k become nonzero near the boundary of Ω_0 and Ω_1 and both y_k and z_k become nonzero near the boundary of Ω_1 and Ω_2 .

Again we try to minimize $\langle\psi|H|\psi\rangle$. For all $k > 0$, $p > 0$, and $k \neq p$,

$$\langle\psi|n_{k\sigma}|\psi\rangle = |y_k|^2 + \frac{|z_k|^2}{2} \quad (72)$$

$$\langle\psi|n_{k\uparrow}n_{k\downarrow}|\psi\rangle = |y_k|^2 \quad (73)$$

$$\langle\psi|b_k^\dagger b_k|\psi\rangle = |y_k|^2 + \frac{|z_k|^2}{2} \quad (74)$$

$$\langle\psi|b_k^\dagger b_{-k}|\psi\rangle = \frac{|z_k|^2}{2} \quad (75)$$

$$\langle\psi|b_k|\psi\rangle = \frac{1}{\sqrt{2}}(x_k^* z_k + z_k^* y_k) \quad (76)$$

$$\langle\psi|b_k^\dagger b_p|\psi\rangle = \frac{1}{2}(z_k^* x_k + y_k^* z_k)(x_p^* z_p + z_p^* y_p). \quad (77)$$

The same equations apply if we take $k \rightarrow -k$, $p \rightarrow -p$ on the left hand sides. Combining everything, and ignoring terms like $g' \sum_k \dots$ that do not scale extensively in the

thermodynamic limit,

$$\langle \psi | H | \psi \rangle = \sum_{k>0} \xi_k \left(4|y_k|^2 + 2|z_k|^2 \right) + U \left(2|y_k|^2 \right) \quad (78)$$

$$- g' \sum_{k,p>0; k \neq p} 2(z_k^* x_k + y_k^* z_k) (x_p^* z_p + z_p^* y_p) \quad (79)$$

$$= \sum_{k>0} (4\xi_k + 2U) |y_k|^2 + 2\xi_k |z_k|^2 \quad (80)$$

$$- 2g' \sum_{k,p>0; k \neq p} (z_k^* x_k + y_k^* z_k) (x_p^* z_p + z_p^* y_p) \quad (81)$$

For each k , introduce a lagrange multiplier λ_k to enforce normalization.

$$0 = \frac{\partial}{\partial x_k} \left[\langle \psi | H | \psi \rangle + \lambda_k \left(|x_k|^2 + |y_k|^2 + |z_k|^2 - 1 \right) \right] \quad (82)$$

$$= \lambda_k x_k^* - 2g' z_k^* \sum_{p>0, p \neq k} (x_p^* z_p + z_p^* y_p) \quad (83)$$

$$\lambda_k = 2 \frac{z_k^*}{x_k^*} O, \quad (84)$$

where $O = g' \sum_{p>0} (x_p^* z_p + z_p^* y_p)$ (now including the contribution $p = k$, which is a $\mathcal{O}(1/L^d)$ difference).

$$0 = \frac{\partial}{\partial y_k^*} [\dots] = (4\xi_k + 2U) y_k - 2z_k O + \lambda_k y_k \quad (85)$$

$$= (4\xi_k + 2U) y_k - 2 \left(z_k - \frac{z_k^* y_k}{x_k^*} \right) O \quad (86)$$

$$2\xi_k + U = \left(\frac{z_k}{y_k} - \frac{z_k^*}{x_k^*} \right) O \quad (87)$$

$$0 = \frac{\partial}{\partial z_k^*} [\dots] = 2\xi_k z_k - 2(x_k O + y_k O^*) + \lambda_k z_k \quad (88)$$

$$= 2\xi_k z_k - 2 \left(x_k O + y_k O^* - \frac{|z_k|^2}{x_k^*} O \right) \quad (89)$$

$$\xi_k x_k^* z_k = \left(|x_k|^2 - |z_k|^2 \right) O + x_k^* y_k O^* \quad (90)$$

$$\xi_k = \left(\frac{x_k}{z_k} - \frac{z_k^*}{x_k^*} \right) O + \frac{y_k}{z_k} O^*. \quad (91)$$

In the last lines, we take the limit $L^d \rightarrow \infty$, so we ignore the g' on the LHS of and also replace the sum in O with a sum over all momentum. Subtracting (91) from (87) gives

$$\xi_k + U = \left(\frac{z_k}{y_k} - \frac{x_k}{z_k} \right) O - \frac{y_k}{z_k} O^*. \quad (92)$$

Using $\xi_k^l = \xi_k$ and $\xi_k^u = \xi_k + U$, and assuming everything is real,

$$\xi_k^l = \left(\frac{x_k}{z_k} + \frac{y_k}{z_k} - \frac{z_k}{x_k} \right) O \quad (93)$$

$$\xi_k^u = - \left(\frac{x_k}{z_k} + \frac{y_k}{z_k} - \frac{z_k}{y_k} \right) O. \quad (94)$$

It is straightforward to check for $U = 0$, combining these equations produces exactly the BCS result, even though we started with a more general wavefunction. This system of two equations is possible to solve analytically, but requires finding the roots of a quartic equation.

c. *Weak coupling* $g \ll U$ and $g \ll W$. First, rewrite

$$\xi_k^l x_k z_k = (x_k^2 - z_k^2 + x_k y_k) O \quad (95)$$

$$\xi_k^u y_k z_k = (z_k^2 - y_k^2 - x_k y_k) O, \quad (96)$$

which now looks very similar to the BCS case, apart from the $x_k y_k$ terms.

If $g \ll U$, we expect that there are still well defined regions $\Omega_0, \Omega_1, \Omega_2$, such that mixing occurs only between x_k and z_k or between y_k and z_k and never x_k and y_k . Is it safe to drop the $x_k y_k$ terms from (95) and (96)? Consider a k point where $\xi_k^l < \xi_k^u < 0$. Here we expect $1 \approx y_k \gg z_k \gg x_k$. If x_k, y_k, z_k are all positive, (95) can only be satisfied if $z_k^2 \gg x_k y_k$. A similar argument can be made for (96). Therefore we drop $x_k y_k$ from both equations and work with

$$\xi_k^l x_k z_k = (x_k^2 - z_k^2) O \quad (97)$$

$$\xi_k^u y_k z_k = (z_k^2 - y_k^2) O. \quad (98)$$

Change variables to

$$x_k^2 - z_k^2 = \frac{\xi_k^l}{E_k^l} (1 - y_k^2), \quad 2x_k z_k = \frac{\Delta_k^l}{E_k^l} (1 - y_k^2) \quad (99)$$

$$E_k^l = \sqrt{\xi_k^{l^2} + \Delta_k^{l^2}} \quad (100)$$

$$z_k^2 - y_k^2 = \frac{\xi_k^u}{E_k^u} (1 - x_k^2), \quad 2y_k z_k = \frac{\Delta_k^u}{F_k^u} (1 - x_k^2) \quad (101)$$

$$E_k^u = \sqrt{\xi_k^{u^2} + \Delta_k^{u^2}} \quad (102)$$

to get

$$\Delta_k^l = g' \sum_{p>0} \frac{\Delta_p^l}{E_p^l} (1 - y_k^2) + \frac{\Delta_p^u}{E_p^u} (1 - x_k^2) \quad (103)$$

$$\Delta_k^u = g' \sum_{p>0} \frac{\Delta_p^l}{E_p^l} (1 - y_k^2) + \frac{\Delta_p^u}{E_p^u} (1 - x_k^2) \quad (104)$$

from which we see that there is only a single momentum-independent parameter Δ defined by

$$1 = g' \sum_{k>0} \frac{1 - y_k^2}{\sqrt{\xi_k^{l^2} + \Delta^2}} + \frac{1 - x_k^2}{\sqrt{\xi_k^{u^2} + \Delta^2}} \quad (105)$$

$$1 = \frac{g}{2} \int d\omega \frac{N''(\omega)}{\sqrt{\omega^2 + \Delta^2}}. \quad (106)$$

This is the same as the BCS gap equation, but with an effective density of states

$$N''(\omega) = \frac{1}{L^d} \sum_k \delta(\omega - \xi_k^l)(1 - y_k^2) + \delta(\omega - \xi_k^u)(1 - x_k^2) \quad (107)$$

Because we are considering $g \ll U$, to a very good approximation $1 - y_k^2 = \theta(\xi_k^u)$ and $1 - x_k^2 = \theta(-\xi_k^l)$.

$$N''(\omega) = \frac{1}{L^d} \sum_k \delta(\omega - \xi_k^l) \theta(\xi_k^u) + \delta(\omega - \xi_k^u) \theta(-\xi_k^l) \quad (108)$$

Note that $N''(\omega)$ is *not* the single-particle density of states of the HK model. In fact it is larger than or equal to it for all ω , and $\int d\omega N''(\omega) \geq 1$. (108) may be rewritten as

$$N''(\omega) = \sum_{k \in \Omega_0} \delta(\omega - \xi_k^l) + \sum_{k \in \Omega_2} \delta(\omega - \xi_k^u) \quad (109)$$

$$+ \sum_{k \in \Omega_1} \delta(\omega - \xi_k^l) + \delta(\omega - \xi_k^u). \quad (110)$$

Mean-field calculation of spectral function

To calculate the spectral function in Fig. 1, we treat the pairing interaction at the mean-field level and the Mott (U) interaction exactly.

$$H_{pair} = -g' \sum_{kk'} b_k^\dagger b_{k'} \quad (111)$$

$$= -g' \sum_{kk'} \langle b_k^\dagger \rangle b_{k'} + b_k^\dagger \langle b_{k'} \rangle \quad (112)$$

$$+ (b_k^\dagger - \langle b_k^\dagger \rangle)(b_{k'} - \langle b_{k'} \rangle) - \langle b_k^\dagger \rangle \langle b_{k'} \rangle. \quad (113)$$

In the mean-field approximation, the third term (quadratic in fluctuations) is dropped. The last term is an inconsequential constant. Define

$$\Delta = -g' \sum_k \langle b_k \rangle \quad (114)$$

$$H_{pair}^{MF} = \sum_k \Delta^* b_k + \Delta b_k^\dagger \quad (115)$$

The full Hamiltonian becomes

$$H^{MF} = \sum_k \xi_k (n_{k\uparrow} + n_{k\downarrow}) + U n_{k\uparrow} n_{k\downarrow} + \Delta^* b_k + \Delta b_k^\dagger. \quad (116)$$

Here, the pairing is treated at the mean-field level but the Mott interaction is treated exactly. While the Hamiltonian no longer separates completely in k -space, each k couples only to $-k$.

$$H^{MF} = \sum_{k>0} H_k^{MF} \quad (117)$$

$$H_k^{MF} = \xi_k (n_{k\uparrow} + n_{k\downarrow} + n_{-k\uparrow} + n_{-k\downarrow}) \quad (118)$$

$$+ U (n_{k\uparrow} n_{k\downarrow} + n_{-k\uparrow} n_{-k\downarrow}) \quad (119)$$

$$+ \Delta^* (b_k + b_{-k}) + \Delta (b_k^\dagger + b_{-k}^\dagger). \quad (120)$$

H^{MF} may be solved by exact diagonalization of each H_k^{MF} , yielding 16 eigenstates and energies for each k . Δ is adjusted for self-consistency such that Eq. 114 is satisfied. The single-particle spectral function is calculated directly from its spectral representation

$$A(k, \omega) = \sum_{nm} |\langle n | c_k | m \rangle|^2 (\rho_n + \rho_m) \delta(\omega + E_n - E_m), \quad (121)$$

where $|n\rangle$ and $|m\rangle$ are eigenstates of H_k^{MF} , $\rho_n = e^{-\beta E_n} / Z$, and the partition function $Z = \sum_n e^{-\beta E_n}$.

Calculation of superfluid stiffness

The superfluid stiffness can be calculated as

$$\frac{D_s}{\pi} = \frac{1}{L^d} \left(\langle K_{xx} \rangle - \int_0^\beta d\tau \langle J_x(\tau) J_x \rangle \right), \quad (122)$$

where

$$K_{xx} = \sum_{k\sigma} \frac{\partial^2 \epsilon_k}{\partial k_x^2} c_{k\sigma}^\dagger c_{k\sigma} \quad (123)$$

$$J_x = \sum_{k\sigma} \frac{\partial \epsilon_k}{\partial k_x} c_{k\sigma}^\dagger c_{k\sigma}. \quad (124)$$

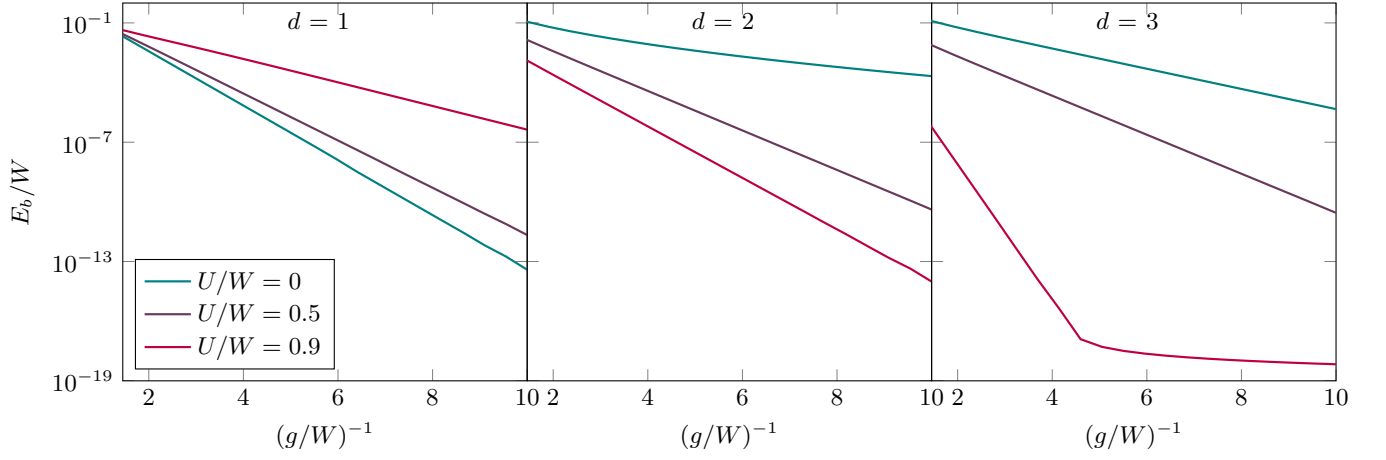
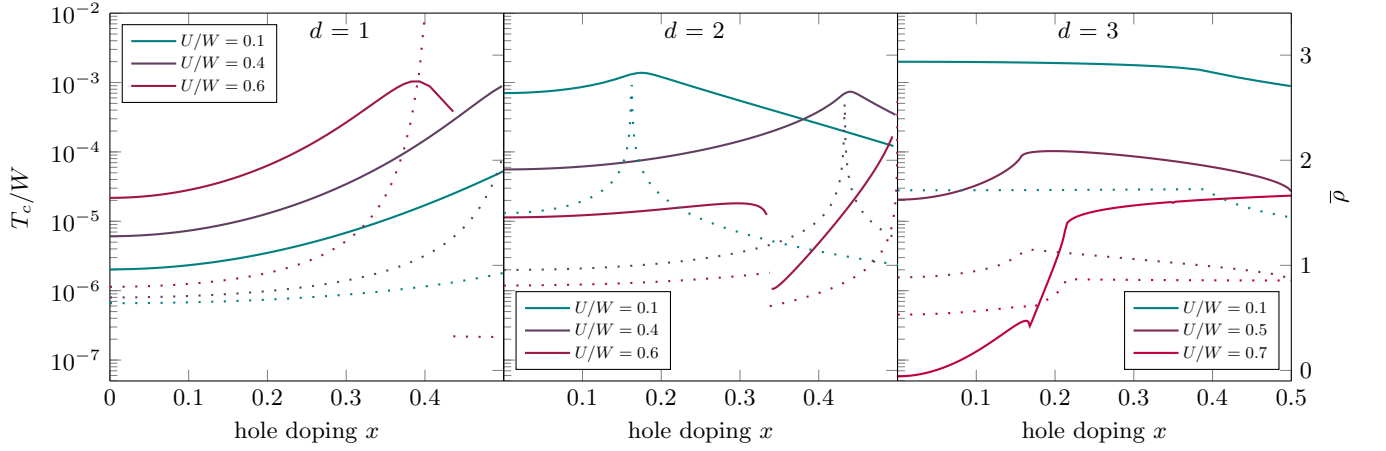
As for the spectral function, all expectation values are calculated by exact diagonalization of H_k^{MF} . In Fig. 4, a 64×64 grid of k -points is used.

For a nearest neighbor tight-binding band structure as considered throughout this work, $\frac{\partial^2 \epsilon_k}{\partial k_x^2} = 2t \cos k_x$, in units where the lattice constant $a = 1$. Therefore $\langle K_{xx} \rangle$ is simply the (negative) kinetic energy along the x -direction bonds. In the spectral representation, we see that

$$\int_0^\beta d\tau \langle J_x(\tau) J_x \rangle = \sum_{nm} |\langle n | J_x | m \rangle|^2 \frac{\rho_m - \rho_n}{E_n - E_m} \quad (125)$$

is nonnegative, so $\frac{\pi}{L^d} \langle K_{xx} \rangle$ is an upper bound to the superfluid stiffness³⁰.

Superconducting energy scales in $d = 1, 2, 3$

FIG. S1. Cooper pair binding energy E_b at half-filling.FIG. S2. Superconducting temperature T_c (solid) and mean density of states $\bar{\rho} = \frac{1}{2}(\rho(\mu) + \rho(\mu - U))$ (dotted) at pair coupling $g/W = 0.1$.